

PERIODISATION MODELS: IN SEARCH OF OPTIMAL PEAK FORM



How was Usain Bolt able to hit peak physical, technical and mental form at 9.35 p.m. on 16 August 2009 at the World Athletics Championships in Berlin, when he ran the 100 metres in 9.58 seconds? Art? Science? Luck? Experience?

The planning of sports training is 10 per cent science and 90 per cent art – but don't turn your nose up at this 10 per cent! Compared to training with no plan at all, a well-designed training plan can improve performance by 1.5 to 2.3 per cent (Muñoz, 2017). Although it might seem trivial, you'll soon see the difference that this can make. A proper training plan also shows greater awareness of the biodynamic changes in your body, as governed by the

principles of training, which translates into a lower chance of injury and, above all, a much greater respect for you and your body.

BASIC CONCEPTS: MACROCYCLE, MESOCYCLE AND MICROCYCLE

Before starting to cook (planning), you need to know what ingredients to use (skills to develop) and what utensils (units of planning) you're going to need.

- **Macrocycle:** the basic unit of planning. A macrocycle is defined as an extended period of training resulting in a higher level of performance (adapted from Solé, 2006). A macrocycle is divided into various mesocycles. It's important to stress that in order to improve performance, it's essential that the body can establish a state of homeostasis.
- **Mesocycle:** a medium-length block of training designed to achieve intermediate goals as part of the wider training process (Solé, 2006). A mesocycle is divided into various microcycles.
- **Microcycle:** a short block of training designed to work on individual skills or qualities.

Defining these three units as long, medium and short blocks of time avoids labelling the macrocycle as a year, the mesocycle as a month or a quarter, and the microcycle as a week. They may, or may not, coincide. You could have several macrocycles in a season, or just one. It all depends on which periodisation model you're using.

There are many periodisation models (linear, non-linear, block, integrated, and so on), each with their own advantages and disadvantages. The suitability of any given model depends on the sport, the athlete and their level of experience. This book will focus on two linear models (the most simple to understand and put into practice) and one block (ATR) model that allows for multiple points of peak form in a season, which, in my experience, is highly applicable and effective for climbing.

As these models are designed for use in (practically) any sport, they refer to periods of training for competitions. However, this doesn't mean they're only designed for competition climbers: simply take competition to mean any challenge, test, checkpoint or project.

LINEAR PERIODISATION: TRADITIONAL AND REVERSE. GENERAL AND SPECIFIC PREPARATION PERIOD. COMPETITION AND TAPERING PERIOD. ADVANTAGES AND DISADVANTAGES. EXAMPLE PERIODISATION

The classic periodisation model, referred to as traditional or linear periodisation, was developed by Matveyev (1977).

It uses the season as the basic unit of planning (the macrocycle). The end of the macro-cycle and the point of peak form coincide with the final or most important competition of the season, although it is possible to have two points of peak form in the same macrocycle.

Matveyev divides the macrocycle into three periods:

- **Base or preparation period:** the aim of this period is to build up basic fitness and movement. It is further divided into:
 1. *General preparation period:* multilateral training (non-specific, varied, multicomponent) takes precedence over specific training (training specifically for climbing). The focus is on volume rather than intensity, creating a solid foundation for the training load in the specific preparation period. It's commonly referred to as the 'preseason'. The mesocycles in this period develop adaptation to strength, endurance and hypertrophy training, through general and non-specific exercises. In training for climbing, this phase should be equally non-specific. We should aim to introduce different types of movement, correct basic technique and amass a broad range of climbing experience (different disciplines, rock types, styles – slabs, cracks, overhangs, dihedrals, and so on), in addition to building continuity and endurance. The general preparation period could account for as little as 20 to 30 per cent of the base period for advanced climbers or as much as 60 to 70 per cent for less advanced climbers.
 2. *Specific preparation period:* specific training takes precedence over multilateral training. There's a notable increase in intensity and a slight reduction in volume. The mesocycles in this phase cover maximum strength, power and RFD. The climbing becomes more specific, applying and developing the skills introduced in the general period in order to maximise performance. The focus shifts to climbing on routes/boulder problems and rock types that are more specific to our end goal for the season.
- **Competition period:** in this period, which is also known as tapering, there's a significant drop in volume and the aim is to reach peak form for the final competition or end goal of the season. There's also a significant drop in specific physical conditioning (we should have already reached maximum potential in the specific preparation period and the goal now is to maintain it). Finally, the focus shifts to specific wall-based training, on either rock or plastic. We should be working at maximum intensity (no higher than in the specific preparation period) on routes where we can perfect the style, grade and skills (strength, continuity, endurance) of our end goal. According to Mujika and Bosquet (2016), tapering is beneficial at any level of ability whenever the training load needs to be reduced.



TAKEN FROM MUJIKA AND BOSQUET (2016)

Any training load will cause some level of fatigue which, logically, will limit our level of performance (the more tired we are, the worse we perform). However, any training load will also raise our physical fitness and, as a result, will improve our level of performance. The success of tapering lies in finding the right balance between our level of fatigue (on a macrocycle level) and our level of physical fitness.

The average improvement in performance observed in quality research is 1.96 per cent. This might seem negligible or even laughable, but according to Mujika and Bosquet (2016):

- In a swimming event at the Sydney 2000 Olympic Games, the difference between first and fourth place was 1.62 per cent, while the difference between third and eighth place was 2.02 per cent.
- In the 1,500 metres at the Athens 2004 Olympic Games, the difference between first and second place was 0.012 seconds (0.05 per cent) and the difference between first and third place was 0.05 seconds (0.23 per cent).

As you can see, at an elite level, margins that might seem trivial are the difference between one medal or another. In climbing, although there are no similar studies, these small margins could make all the difference when it comes to sticking a hard move or using a crucial handhold.

Mujika and Bosquet (2016) give the following guidelines for tapering:

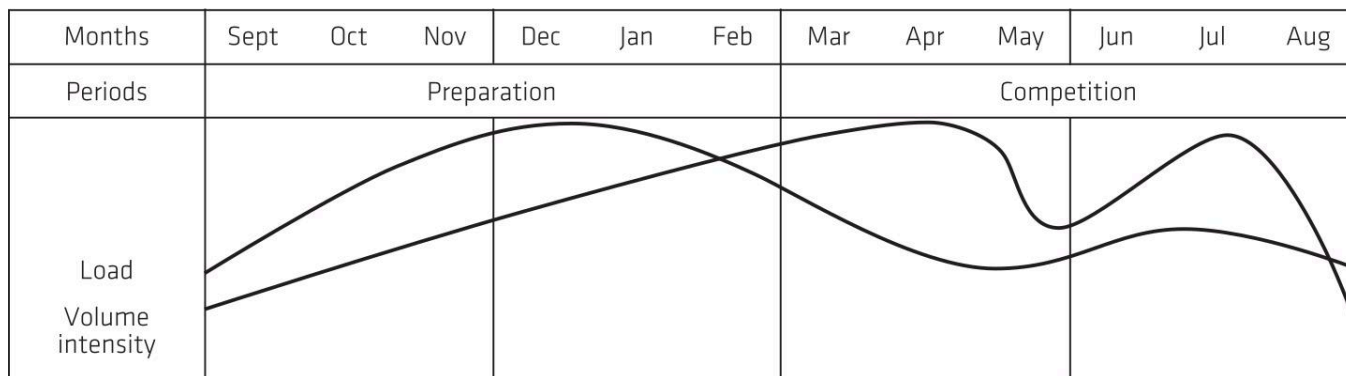
- The volume of training should be reduced by 60 to 41 per cent over the four weeks prior to the competition.
- First reduce the number of training sessions, then reduce the duration of the sessions.
- The intensity should remain constant, not increase.
- The last few sessions shouldn't be used as a test.
- Calorie intake should be adapted to the new energy needs in order to avoid weight gain.
- Priority should be given to movement efficiency and the development of maximum power.
- For best results, favour heavy weights over light weights.
- It's essential to ensure full recovery.
- When done correctly, tapering increases the diameter of fast-twitch muscle fibres, boosts neural electrical activity and makes glycolysis more efficient at maximum intensity, in addition to modifying anabolic hormonal response.

Mujika and Bosquet refer to a quote from the world of competitive swimming, which we could all bear in mind when tapering or approaching our end goal(s): 'You don't need to feel good to swim fast.' Feelings are fundamental, especially when we're facing a win-or-lose situation after months and months of training. However, if we've trained properly and our body has adapted to the training load and is in a state of homeostasis, we're much more likely to top our target route, even if we start doubting our chances of success.

- **Transition period:** this is when we start losing form and begin the process of recovery and regeneration.

Micro 1 2 3	Micro 1 2 3	Micro 1 2 3	Micro 1 2 3	Micro 1 2 3	Micro 1 2 3	Period of peak form
Meso 1	Meso 2	Meso 3	Meso 4	Meso 5	Meso 6	
General period			Specific period		Tapering	
Preparation period					Competition period	
Macrocycle						

This model of planning is known as linear periodisation as there is a linear or proportional increase in intensity and a linear or proportional decrease in volume over the course of the macrocycle.



What are the advantages and disadvantages of Matveyev's linear periodisation model?

Main advantages of linear periodisation:

- It lays strong foundations in each skill or quality as they are trained over a long period of time.
- The risk of injury is fairly low as it's a progressive system.
- It's a safe model for beginners.

Main disadvantages of linear periodisation:

- It takes a long time to reach peak form.
- There's no scope for readjustment until the first period of peak form.
- It's not recommended for advanced athletes as the skills aren't revisited during the year.
- It's fallen into disuse in many sports as it's less applicable to performance.

The following is an example of linear periodisation for a novice sport climber looking to up their grade from the F5s to F6s for a sport climbing trip in the Easter holidays, beginning their macrocycle at the start of September. It should give an idea of the nature of training in each period and mesocycle.

- **Macrocycle: September–April**

- **Preparation period: 1 September to 24 March**

- **General preparation period: 1 September to 23 December**

- **Mesocycle 1: 1 to 30 September (4 weeks)**

Physical conditioning: strength training circuit including exercises for all muscle groups. Level of intensity: EL 8(20). Low-intensity steady-state aerobic endurance – working in Z1 (130 to 155bpm) for a total of 2 hours per week.

Climbing: long and easy traverses (below maximum grade), with short rests. Focus on building up the number of moves, not the difficulty. Easy boulder problems.

- **Mesocycle 2: 1 October to 11 November (6 weeks)**

Physical conditioning: hypertrophy training. Level of intensity: EL 8(16). Special focus on core and shoulders. Low-intensity steady-state aerobic endurance – working in Z1 for a total of 1 hour 45 minutes per week.

Sprint interval training: 2 minutes of effective work per session.

Climbing: work on movement and technique, focusing on bicycles, foot swaps and movement efficiency. Endurance training – intermittent method, aiming for up to 12 minutes of effective work. Maximum pulling strength: EL 8(16).

- **Mesocycle 3: 12 November to 23 December (6 weeks)**

Physical conditioning: hypertrophy training. Level of intensity: EL 6(12). Low-intensity steady-state aerobic endurance – working in Z1 for a total of 45 minutes per week and in Z3 (160 to 170bpm) for 30 minutes per week.

Sprint interval training: 2 minutes 30 seconds of effective work per session.

Climbing: work on movement and technique, introducing overhangs, dragging and heel hooks. Endurance training – long interval method, aiming for up to 20 minutes of effective work. Maximum pulling strength, intensity: EL 6(12).

- **Specific preparation period: 7 January to 17 March**

- **Mesocycle 4: 7 January to 17 February (6 weeks)**

Physical conditioning: maximum strength training, multi-joint exercises only, special focus on pulling in different grip positions. Maximum intensity method II. SIT, with a total of 3 minutes of effective work.

Climbing: endurance training. Continuous method, introducing active wall-based recovery. Maximum grip strength training, aiming to hang for 1 minute 30 seconds on a pull-up bar.

- **Mesocycle 5: 18 February to 24 March (5 weeks)**

Physical conditioning: maximum strength training, multi-joint exercises only, special focus on pulling in different grip positions and upper body pushing strength. Maximum intensity method II. SIT, with a total of 3 minutes of effective work.

Climbing: endurance training. High-intensity interval method with very short intervals (three to six moves) and easy climbing between intervals on a vertical wall for as long as it takes to fully recover (active wall-based recovery). Maximum grip strength training, aiming to hang for 1 minute 30 seconds on a fingerboard, using a three-pad hold.

- **Competition period: 25 March to 21 April**

- **Tapering: 25 March to 14 April (3 weeks)**

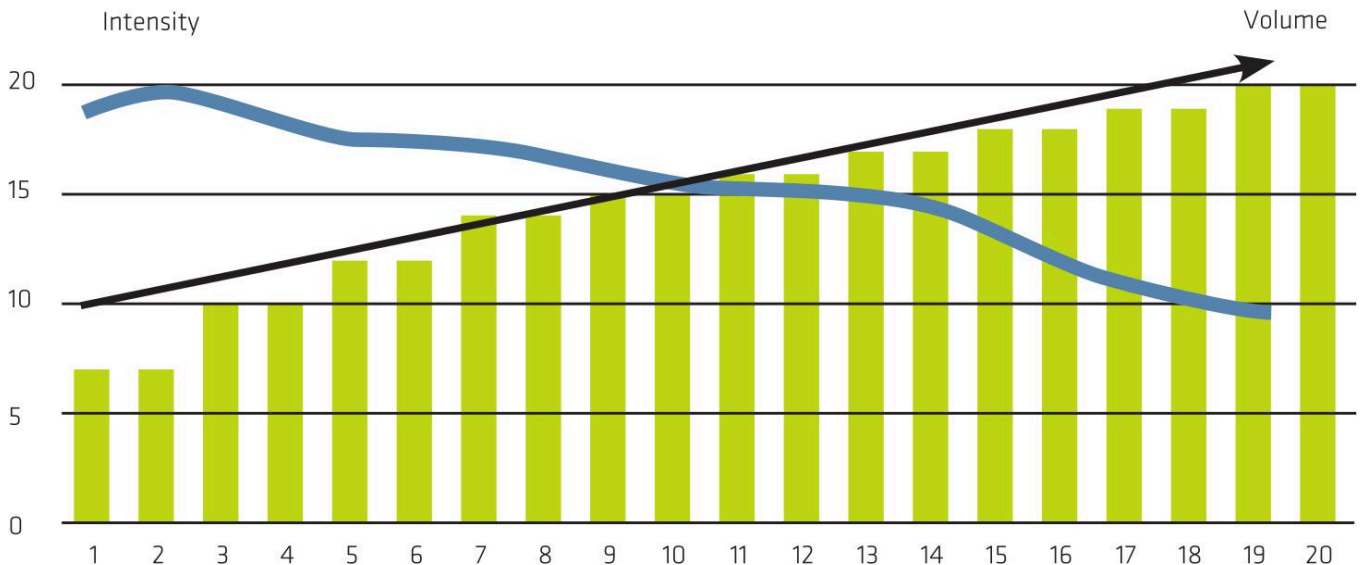
Physical conditioning: none.

Climbing: endurance training, interval method with short intervals (six to eight moves) on a slightly overhanging wall and easy climbing between intervals on a vertical wall for as long as it takes to fully recover (active wall-based recovery). Maximum pulling strength and maximum grip strength.

- **Peak form: 15 to 21 April (1 week)**

Sport climbing trip.

There's another linear model, known as reverse periodisation, which was developed by Ian King (2000). The main difference between the reverse model and Matveyev's model is that the levels of volume and intensity are reversed.



As seen in the image, and in contrast to linear periodisation, King suggests starting at maximum intensity and minimum volume, then gradually reducing the intensity and increasing the volume over the course of the season. That is, in the preparation period, you'd begin by training maximum strength and power, then move on to endurance. According to House and Johnston (2014), reverse periodisation is the best training method for climbers and mountaineers. This is because it begins by recruiting the greatest number of fast-twitch muscle fibres, through maximum strength (high-intensity) training, and then converts them into slow-twitch muscle fibres for a greater number of fibres that provide energy for longer periods of time. Reverse periodisation is also recommended by another great mountaineer, Mark Twight, in his book *Extreme Alpinism*.

Main advantages of reverse periodisation:

- Like linear periodisation, it lays strong foundations in each skill or quality as they are trained over a long period of time.
- It's the best option for trainee athletes or athletes with little training experience.
- It ensures a higher degree of fibril recruitment, the prevalence of type II muscle fibres and enhanced mitochondrial density, guaranteeing greater functional capacity.

Main disadvantages of reverse periodisation:

- Like Matveyev's model, it takes a long time to reach peak form.
- There's also no scope for readjustment until the first point of peak form.
- It's also not recommended for advanced athletes as the skills aren't revisited during the year.

The following is an example of reverse periodisation for an intermediate sport climber looking to consolidate at F6a to F6c and start pushing into the F7s. It should give an idea of the nature of training in each period and mesocycle.

- **Macrocycle: September–March**

- **Preparation period: 1 September to 24 February**

- **General preparation period: 1 September to 2 December**

- **Mesocycle 1: 1 to 16 September (2 weeks)**

Physical conditioning: strength training circuit including exercises for all muscle groups. Level of intensity: EL 8(20). Low-intensity steady-state aerobic endurance – working in Z1 (130 to 155bpm) for a total of 1 hour per week.

Climbing: long and easy traverses (at least one and a half to two grades below maximum grade),

with short rests. Focus on building up the number of moves and the range of techniques, not the difficulty.

- **Mesocycle 2: 17 September to 21 October (5 weeks)**

Physical conditioning: maximum strength training. Level of intensity: EL 4(7). Low-intensity steady-state aerobic endurance – working in Z1 for a total of 45 minutes per week.

Sprint interval training: 2 minutes of effective work per session.

Climbing: bouldering. Endurance training – continuous method, aiming for up to 30 minutes of effective work. Maximum grip strength – max weight deadhangs on a large edge.

- **Mesocycle 3: 22 October to 2 December (6 weeks)**

Physical conditioning: maximum strength training. Level of intensity: EL 4(7).

Sprint interval training: 2 minutes of effective work per session.

Climbing: endurance training – long interval method, aiming for up to 40 minutes of effective work. Maximum grip strength – minimum edge deadhangs and lock-offs. Power training and RFD on the campus board.

- **Specific preparation period: 3 December to 24 February**

- **Mesocycle 4: 3 December to 13 January (6 weeks)**

Physical conditioning: strength training with an emphasis on single-joint upper body exercises, intensity EL 8(18).

Climbing: endurance training. Continuous method on an overhanging wall, with active wall-based recovery.

- **Mesocycle 5: 14 January to 24 February (6 weeks)**

Physical conditioning: strength training with an emphasis on single-joint upper body exercises, intensity EL 8(18).

Climbing: endurance training. Interval method with medium intervals (eight to ten moves) on an overhanging wall and easy climbing between intervals on an overhanging wall for as long as it takes to fully recover (active wall-based recovery). Intermittent deadhangs.

- **Competition period: 25 February to 24 March**

- **Tapering: 25 February to 17 March (3 weeks)**

Physical conditioning: none.

Climbing: endurance training, continuous method on an overhanging wall, aiming for up to 30 minutes of effective work. Maximum pulling strength and maximum grip strength – max weight deadhangs on a large edge.

- **Peak form: 18 to 24 March (1 week)**

Begin working routes graded F7a and above.

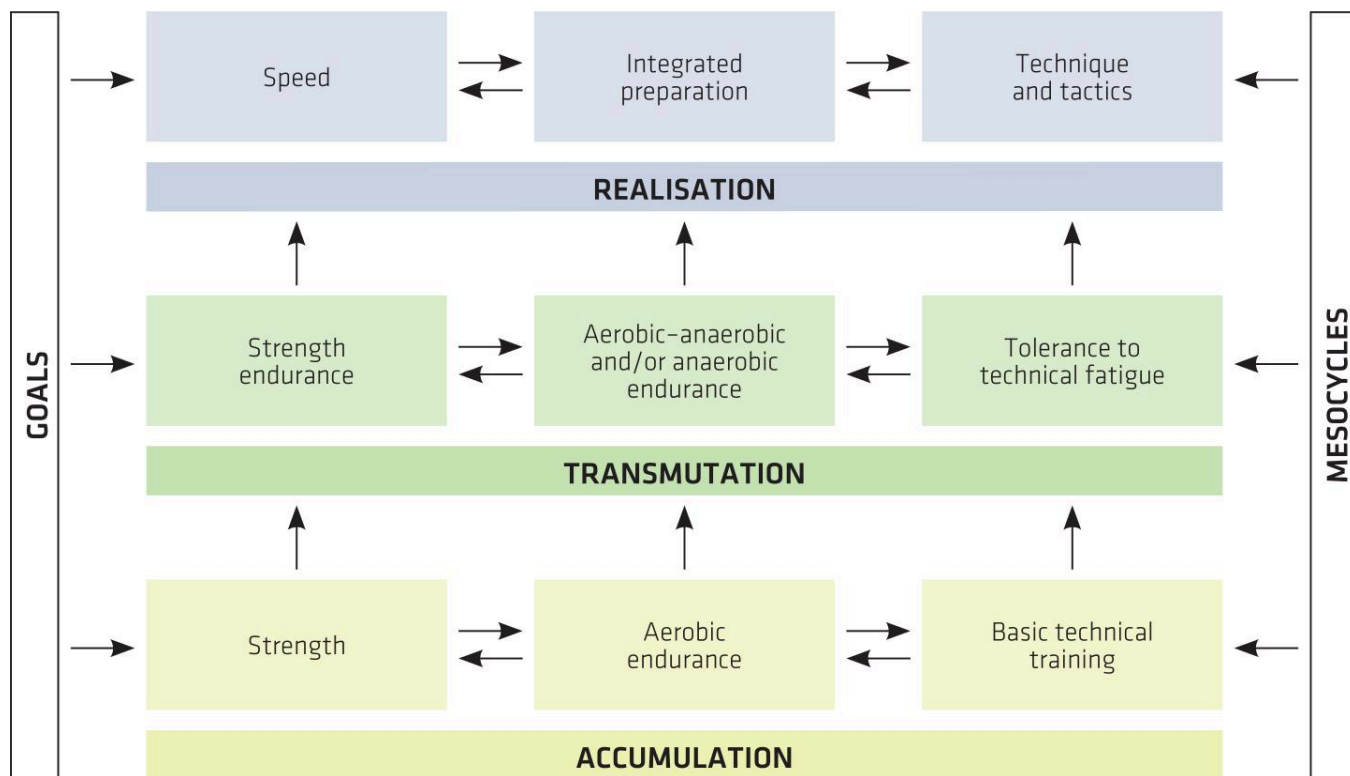
ATR PERIODISATION. ACCUMULATION, TRANSMUTATION AND REALISATION MESOCYCLES. TYPES AND ORDER OF MICROCYCLES. ADVANTAGES AND DISADVANTAGES. EXAMPLE PERIODISATION

The ATR model is one of the newer forms of periodisation and is based on the idea of training blocks. It was created by Issurin and Kaverin (1985) and developed by Fernando Navarro Valdivielso. It's the method used by top Spanish climber and coach Patxi Usobiaga, who has coached Adam Ondra, among others.

Each macrocycle is six to nine weeks long, meaning peak form can be achieved in a short period of time and multiple times across the season.

Each macrocycle comprises three mesocycles: accumulation, transmutation and realisation.

- *Accumulation:* this mesocycle lasts for two to three weeks. The aim is to build the athlete's technical and movement potential, which is similar to the general preparation period in linear periodisation; the technical and movement skills accumulated in this mesocycle would create a base for the specific preparation period. Accumulation covers basic endurance, basic strength and basic technique, focusing on the correction of errors. In climbing, this includes pulling strength, continuity, general increase in ROM and a range of non-specific boulder problems and routes that aren't too narrowly focused on the end goal. The volume of training is relatively high, with a moderate level of intensity.
- *Transmutation:* this mesocycle lasts for two to three weeks. The aim is to transform the athlete's technical and movement potential into specific skills, turning basic movement and technique into sport-specific abilities. Transmutation covers specific strength, specific endurance and specific technique in situations of fatigue. In climbing, this includes finger strength endurance, power and RFD, as well as endurance. This mesocycle places special emphasis on tolerance to fatigue and retaining good technique when fatigued. The training volume is lower than in the accumulation phase and the training intensity is high. It is important to be well rested for the training in this mesocycle.
- *Realisation:* this mesocycle lasts for two to three weeks. The aim is to achieve the best possible competition results by employing to the fullest possible extent all technical and movement skills developed in the previous cycles in the specific competition activity. The training involves competition-style exercises – boulder problems and routes that are specific to the end goal, maintaining maximum intensity, consolidating at the target grade and ensuring being properly rested for training. To achieve supercompensation, it also includes recovery protocols and tapering. At the end of this block, you'd look to achieve the end goal of the macrocycle.

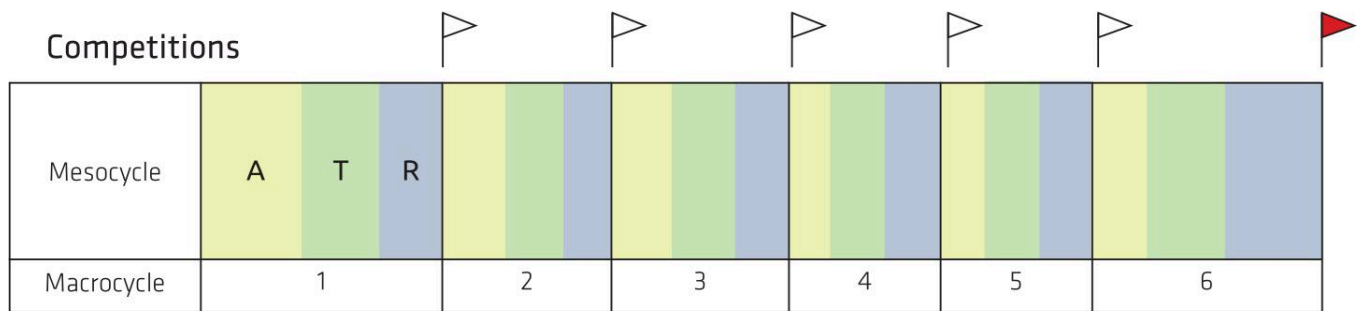


There are several types of microcycle for each mesocycle, and the microcycles that you might do would depend on where you are in the season, your specific goals, the level of physical/technical demand, your level of ability, and so on.

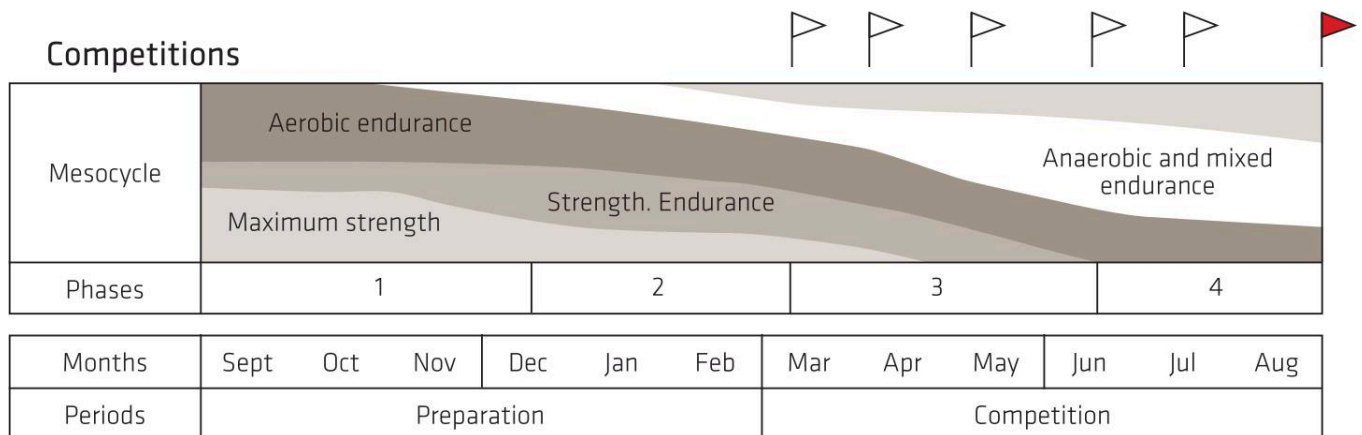
Accumulation	Transmutation	Realisation
Recovery and/or adjustment	Adjustment and load	Adjustment or load
Load and/or impact	Load and/or impact	Load or recovery
Load	Load and/or impact	Activation and competition

Here's a description of what each microcycle entails:

- **Adjustment:** medium training load. Medium intensity. Medium technical difficulty. Duration: 4 to 7 days.
- **Load:** *significant* load and total workload. Duration: 1 week.
- **Impact:** maximum total workload. Load increased to *extreme* limits. Duration: 1 week.
- **Activation:** Low volume. Increased intensity. Modelling of upcoming competitions. Duration: 3 to 7 days.
- **Competition:** competition schedule and recovery protocols. Duration: 3 to 9 days.
- **Recovery:** low volume and intensity. Varied means of recovery. You'd use this microcycle when the previous macrocycle ends with a significant load and/or biological stress. Duration: 3 to 7 days.



A – Accumulation. T – Transmutation. R – Realisation



This graphic gives a comparison between reverse linear periodisation and ATR block periodisation.

Inevitably, there are also advantages and disadvantages to the ATR model.

Main advantages of ATR periodisation:

- It allows for multiple points of peak form and training adjustments throughout the season.
- Peak form can be reached in a very short period of time.
- It's the best option for high-level athletes with various competitions across the season.
- It helps keep the athlete motivated as the content of the training is very varied.

Main disadvantages of ATR periodisation:

- It's a complex model that's very time consuming for the coach.
- It doesn't allow any time to consolidate or stabilise the targeted skills.

The following is an example of ATR periodisation for an advanced sport climber looking to climb a F7c route that they've been working for some time. It should give an idea of the nature of training in each period and mesocycle.

- **Macrocycle: March to April**
- **Accumulation mesocycle: 4 March to 21 March**
 - **Adjustment microcycle 1: 4 March to 7 March (4 days)**
Physical conditioning: maximum strength, multi-joint exercises, intensity EL 1(4).
Climbing: emphasis on technique. Bouldering.
 - **Load microcycle 1: 8 March to 14 March (1 week)**
Physical conditioning: maximum strength, multi-joint exercises, intensity EL 2(3).
Climbing: bouldering. Maximum grip strength – max weight deadhangs on a large edge.
 - **Load microcycle 2: 15 March to 21 March (1 week)**
Physical conditioning: maximum strength, multi-joint exercises, intensity EL 2(3).
Climbing: bouldering. Power training and RFD on the campus board.
- **Transmutation mesocycle: 22 March to 11 April**
 - **Load microcycle 3: 22 March to 28 March (1 week)**
Climbing: endurance training. Continuous method with blood flow restriction, aiming for 20 minutes of climbing. Continue climbing with no blood flow restriction, using the long interval method, for a total of 50 minutes of effective work.
 - **Impact microcycle 1: 29 March to 4 April (1 week)**
Climbing: intermittent deadhangs. Endurance training. Interval method with short intervals (six to eight moves), difficulty F7a/7a+, on an overhanging wall, and active wall-based recovery between intervals for 2 minutes, also on an overhanging wall, difficulty F6a/6b.
 - **Impact microcycle 2: 5 April to 11 April (1 week)**
Climbing: intermittent deadhangs. Endurance training. Interval method with medium intervals (ten to

twelve moves), difficulty F7a/7a+, on an overhanging wall, and active wall-based recovery between intervals for 1 minute 30 seconds, also on an overhanging wall, difficulty F6a/6b.

- **Realisation mesocycle: 12 April to 21 April**

- **Adjustment microcycle 2: 12 April to 15 April (4 days)**

- Climbing:* bouldering. Maximum grip strength – max weight deadhangs on a large edge.

- **Activation microcycle: 16 April to 18 April (3 days)**

- Climbing:* bouldering. Maximum pulling strength, intensity EL 1(3) and maximum grip strength – minimum edge.

- **Competition microcycle: 19 April to 21 April (3 days)**

- Work on target route.